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Diagnostic laboratory systems and methods of imaging tube assemblies

Abstract

A method of synthesizing an image of a tube assembly includes capturing an image of the tube assembly, wherein the capturing generates a captured image. The captured image is decomposed into a plurality of features in latent space using a trained image decomposition model. One or more of the features in the latent space is manipulated into one or more manipulated features. A synthesized tube assembly image is generated with at least one of the manipulated features using a trained image composition model. Other methods and systems are disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a 35 U.S.C. 371 national application of international application no. PCT/US2023/063622 filed Mar. 2, 2023, which claims priority to U.S. Provisional application No. 63/268,846 filed Mar. 3, 2022, the contents of which are fully incorporated herein by reference.

FIELD
(1) Embodiments of the present disclosure relate to diagnostic laboratory systems and methods of imaging tube assemblies in diagnostic laboratory systems.
BACKGROUND
(2) Diagnostic laboratory systems conduct clinical chemistry or assays to identify analytes or other constituents in biological samples such as blood serum, blood plasma, urine, interstitial liquid, cerebrospinal liquids, and the like. The samples may be received in and/or transported throughout a system in sample tube assemblies.
(3) Many of the laboratory systems process large volumes of tube assemblies and the samples in the tube assemblies. Some laboratory systems make use of machine vision and machine learning to

facilitate sample processing and tube assembly identification (e.g., characterization). For example, vision-based machine learning models (e.g., AI models) have been adopted to provide fast and noninvasive methods for tube assembly identification and fluid characterization. However, the training cost for adding new tube assemblies may be excessive because large amounts of training data may be required to retrain or adapt the machine learning models to characterize the new tube assemblies.

(4) Tube assembly manufacturers continuously produce new tube assembly configurations to either bring in new features or save production costs. In addition, third-party low-cost tube assembly replacements may be utilized by some laboratories due to availability and/or financial reasons. With so many tube assembly configurations, it is difficult to collect every possible tube assembly variation across the globe, which means that the machine learning models likely cannot be trained on all the tube assembly configurations. Accordingly, there is a limit to the number of tube assembly configurations the machine learning models can handle in practice. Therefore, a need exists for laboratory systems and methods that facilitate the introduction of new tube assembly configurations into the laboratory systems.

SUMMARY

(5) According to a first aspect, a method of synthesizing an image of a tube assembly is provided. The method includes capturing an image of a tube assembly, the capturing generating a captured image; decomposing the captured image into a plurality of features in latent space using a trained image decomposition model; manipulating one or more of the features in the latent space into manipulated features; and generating a synthesized tube assembly image with at least one of the manipulated features using a trained image composition model.

(6) In a further aspect, a method of synthesizing images of tube assemblies is provided. The method includes constructing an image decomposition model configured to receive input images of tube assemblies and to decompose the input images to a plurality of features in a latent space; constructing an image composition model configured to compose synthetic tube assembly images based on the plurality of features in the latent space; and training the image decomposition model and the image composition model using at least an image of a first tube assembly and an image of a second tube assembly with one or more known variations between the image of the first tube assembly and the image of the second tube assembly, wherein the training produces a trained image decomposition model and a trained image composition model.

(7) In another aspect, a diagnostic laboratory system is provided. The diagnostic laboratory system includes an image decomposition model configured to receive input images of tube assemblies and to decompose the input images into a plurality of features in a latent space; and an image composition model configured to compose synthetic tube assembly images based on the plurality of features in the latent space, wherein the image decomposition model and the image composition model are trained using at least an image of a first tube assembly and an image of a second tube assembly with one or more known variations between the image of the first tube assembly and the second tube assembly.

(8) Still other aspects, features, and advantages of this disclosure may be readily apparent from the following description and illustration of a number of example embodiments, including the best mode contemplated for carrying out the disclosure. This disclosure may also be capable of other and different embodiments, and its several details may be modified in various respects, all without departing from the scope of the disclosure. This disclosure is intended to cover all modifications, equivalents, and alternatives falling within the scope of the claims and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings, described below, are provided for illustrative purposes, and are not necessarily drawn to scale. Accordingly, the drawings and descriptions are to be regarded as illustrative in nature, and not as restrictive. The drawings are not intended to limit the scope of the disclosure in any way.

(2) FIG. **1** illustrates a block diagram of a diagnostic laboratory system including a plurality of instruments according to one or more embodiments.

(3) FIG. **2** illustrates a top plan view of an imaging instrument provided in a diagnostic laboratory system according to one or more embodiments.

(4) FIGS. **3**A-**3**C illustrate different types of tube assemblies including caps affixed to tubes that may be used within a diagnostic laboratory system according to one or more embodiments.

(5) FIGS. **4**A-**4**C illustrate different types of tubes of tube assemblies that may be used within a diagnostic laboratory system according to one or more embodiments.

(6) FIG. **5** illustrates a block diagram of a process that implements a method of synthesizing images of tube assemblies according to one or more embodiments.

(7) FIG. **6** illustrates a block diagram of a process that implements a method of synthesizing images of exteriors and interiors of tube assemblies according to one or more embodiments.

(8) FIG. **7** illustrates a process that may be used to train an image decomposition model and an image composition model according to one or more embodiments.

(9) FIGS. **8**A and **8**B illustrate images of similar tube assemblies, wherein the image of FIG. **8**A has a blue cap and the image of FIG. **8**B has a green cap.

(10) FIG. **8**C illustrates an example of a decomposed image of the tube assembly of FIG. **8**A according to one or more embodiments.

(11) FIGS. **9**A-**9**B illustrate images of tube assemblies having different interior and/or fluid features according to one or more embodiments.

(12) FIG. **10** is a flowchart illustrating a method of synthesizing an image of a tube assembly according to one or more embodiments.

(13) FIG. **11** is a flowchart illustrating another method of synthesizing an image of a tube assembly according to one or more embodiments.

DETAILED DESCRIPTION

(14) Diagnostic laboratory systems conduct clinical chemistry and/or assays to identify analytes or other constituents in biological samples such as blood serum, blood plasma, urine, interstitial liquid, cerebrospinal liquids, and the like. The samples are collected in tube assemblies (e.g., sample tubes) and transported via the tube assemblies to the laboratory systems where the samples are analyzed. In some embodiments, the tube assemblies are transported throughout the laboratory system so that various instruments may perform the analyses.

(15) Laboratory systems may use tube assemblies from various manufacturers to collect samples and transport the samples to and throughout a laboratory system. A tube assembly may include a sample tube, such as a closed-bottomed tube. Some tube assemblies may include a cap that seals the tube. A tube assembly may also include the contents of the tube. Different tube assembly types may have different characteristics, such as different sizes and different chemical additives therein. For example, many tube assembly types are chemically active, meaning the tube assemblies contain one or more additive chemicals that are used to change or retain a state of the samples or otherwise assist in sample processing. In some embodiments, the inside wall of a tube may be coated with the one or more additives, or the additives may be provided elsewhere in the tubes. The types of additives contained in the tubes may be serum separators, coagulants such as thrombin, anticoagulants such as EDTA or sodium citrate, anti-glycosis additives, or other additives for changing or retaining a characteristic of the samples. The tube assembly manufacturers may associate the colors of the caps on the tubes and/or shapes of the tubes or caps with specific types

of chemical additives contained in the tubes.

(16) Different manufacturers may have their own standards for associating attributes of the tube assemblies, such as cap color, cap shape, and tube shape with particular properties of the tube assemblies. For example, the attributes may be related to the contents of the tubes or possibly whether the tubes are provided with vacuum capability. In some embodiments, a manufacturer may associate all tube assemblies with gray colored caps with tubes including potassium oxalate and sodium fluorate configured to test glucose and lactate. Tube assemblies with green colored caps may include heparin for stat electrolytes such as sodium, potassium, chloride, and bicarbonate. Tube assemblies with lavender caps may identify tubes containing EDTA (ethylenediaminetetraacetic acid-an anticoagulant) configured to test CBC w/diff., HgBA1c, and parathyroid hormone. Other cap colors such as red, yellow, light blue, royal blue, pink, orange, and black may be used to signify other additives or lack of an additive. In other embodiments, combinations of colors of the caps may be used, such as yellow and lavender to indicate a combination of EDTA and a gel separator, or green and yellow to indicate lithium heparin and a gel separator.

(17) The laboratory systems may use the tube assembly attributes for further processing of the tubes and/or the samples. Since the tubes may be chemically active, it is important to associate specific tests that can be performed on samples with specific tube assembly types because the tests may be dependent on the contents of the tubes. Thus, the laboratories may confirm that tests being run on samples in the tubes are correct by analyzing the colors and/or shapes of the caps and/or the tubes. Other attributes may also be analyzed.

(18) Transport mechanisms, such as robots and tube assembly carriers within the laboratory systems may have specific hardware and processes for moving different types of tube assemblies. For example, a robot may grasp a first type of tube assembly differently than a second type of tube assembly. In addition, the laboratory systems may utilize different carriers depending on the types of tube assemblies that are to be transported throughout the laboratory systems. Thus, the laboratory systems need to identify the tube assemblies.

(19) The laboratory systems described herein may use vision systems to capture images of tube assemblies and identify the tube assemblies and/or contents of the tube assemblies. For example, the laboratory systems may include vision-based artificial intelligence (AI) models configured to provide fast and noninvasive methods for tube assembly characterization. Embodiments of the AI models may be trained to characterize different types of tube assemblies and slight variations of the tubes and/or the caps. As new types of tube assemblies are introduced into the laboratory systems, the AI models need to be updated to be able to classify the new types of tube assemblies. Retraining the AI models in conventional laboratory systems may be costly and time consuming because a plurality of tube assembly types need to be imaged and manually classified to retrain the AI models.

(20) The systems and methods described herein overcome the problems with tube assembly classification by synthesizing images of tube assemblies. Images of synthetic data and/or paired tube assemblies may be used with controlled variations in the images to train an image decomposition model and an image composition model. The trained decomposition model is used to decompose images of tube assemblies into features (e.g., decomposed features) in latent space. One or more of the features in the latent space may then be manipulated. The trained composition model may reassemble the features, including a manipulated feature, to generate synthesized tube assembly images.

(21) In some embodiments, image-to-image translation is used by the decomposition model and the composition model to synthesize images of the tube assemblies, wherein the new or synthesized images are controlled modifications of reference images. Image-to-image translation is a class of vision and graphics processing wherein the goal is to learn the mapping between an input image of a tube assembly and an output image of a tube assembly using a training set of aligned image pairs.

In some embodiments, the processing may include using a generative adversarial network (GAN) to perform the image-to-image translation. These and other systems and methods are described below in greater detail with reference to FIGS. **1-11**.

(22) Reference is now made to FIG. **1**, which illustrates an example embodiment of an automated diagnostic system **100** configured to process and/or analyze biological samples stored in tube assemblies **102** (e.g., sample containers). The tube assemblies **102** may be any suitable containers (e.g., tubes), including transparent or translucent containers, such as blood collection tubes, test tubes, sample cups, cuvettes, or other containers capable of containing samples and/or allowing imaging of the samples contained therein. The tube assemblies may or may not have caps or lids attached thereto. As described herein, the tube assemblies **102** may have different sizes and may have different cap colors and/or cap types. As described herein, the system **100** and methods described herein enable the system **100** or devices coupled to the system **100** to generate images that synthesize different tube assemblies.

(23) The system **100** may include a plurality of instruments **104** that are configured to process the tube assemblies **102** and/or samples located in the tube assemblies **102**. The tube assemblies **102** may be received at the system **100** at an input/output instrument **104**A that may include on one or more racks **106** and a robot **108**. The robot **108** may transport the tube assemblies **102** between the racks **106** and carriers **114** located on a track **112**. The carriers **114** are configured to transport the tube assemblies **102** on the track **112**.

(24) The instruments **104** may process the samples and/or the tube assemblies **102**. The processing may include preprocessing or prescreening the samples and/or the tube assemblies **102** prior to analysis by one or more of the instruments **104**. For example, the preprocessing may prepare samples for specific assays or other analyses. Preprocessing may also identify the types of tube assemblies **102** to prepare the samples for analyses in response to the identification. Other ones of the instruments **104** may perform the analyses on the samples.

(25) One or more of the instruments **104** may include an imaging device that captures images of the tube assemblies **102** and/or the carriers **114**. In the embodiment of FIG. **1**, an imaging instrument **104**B may include imaging devices as described below and may be configured to capture images of the tube assemblies **102** and/or the carriers **114**. In some embodiments, an imaging device **116** may be located in or proximate the input/output instrument **104**A and may be configured to capture images of the tube assemblies **102** prior to the tube assemblies **102** being transferred to the track **112** or loaded into the carriers **114**.

(26) Additional reference is made to FIG. **2**, which illustrates a top plan view of the imaging instrument **104**B. Other embodiments of imaging instruments may be implemented in the system **100**. In the embodiment of FIG. **2**, the imaging instrument **104**B is configured to capture images of the tube assemblies **102** while the tube assemblies **102** are located on the track **112**. A tube assembly **202** is shown located on the track **112** and at an imaging location **214** within the imaging instrument **104**B. The components of the imaging instrument **104**B may be controlled by a computer **220**. In some embodiments, the computer **220** may be incorporated into the computer **120** (FIG. **1**).

(27) The imaging instrument **104**B may include three imaging devices **216** configured to capture images of the tube assembly **202** at the imaging location **214**. The imaging devices **216** are referred to individually as a first imaging device **216**A, a second imaging device **216**B, and a third imaging device **216**C. In other embodiments, the imaging instrument **104**B may include one or more imaging devices. The imaging devices **216** may be configured to generate image data representative of the tube assembly **202**. The computer **220** and/or the computer **120** (FIG. **1**) may be configured to process the image data generated by the imaging devices **216** as described herein.

(28) The imaging instrument **104**B may also include one or more illumination devices **217** that are configured to illuminate the imaging location **214**. Thus, the illumination devices **217** may be configured to illuminate the tube assembly **202**. In the embodiment of FIG. **2**, the imaging

instrument **104**B may include three illumination devices **217**, which are referred to individually as a first illumination device **217**A, a second illumination device **217**B, and a third illumination device **217**C.
(29) Referring again to FIG. **1**, the system **100** may also include or be coupled to a computer **120** that may be configured to operate and/or communicate with the instruments **104**. In some embodiments, the computer **120** may analyze data received in the system **100** and/or generated by the instruments **104**. The computer **120** and programs executed by the computer **120** may also analyze image data generated by the imaging instrument **104**B and other imaging devices as described herein. The computer **120** may include a processor **122** configured to execute computer-readable instructions (e.g., program code).
(30) In the embodiment of FIG. **1**, the computer **120** may be connected to individual ones of the instruments **104**. For example, the computer **120** may be connected to the computer **220** in the imaging instrument **104**B. In some embodiments, the computer **120** may be coupled to one or more workstations (not shown) coupled to one or more of the instruments **104**. In some embodiments, the computer **120** may be configured to transmit computer instructions and/or computer code to individual ones of the instruments **104**. In some embodiments, the computer **120** may be configured to receive and/or process data generated by the instruments **104**.
(31) The computer **120** may include or be coupled to memory **124** that may store or access one or more modules and/or programs **126**, which are described herein as being stored in the memory **124**. The modules or programs **126** may be executable by the processor **122**. The programs **126** may be configured to operate one or more of the instruments **104**. Although the programs are described as individual programs, in some embodiments the programs **126** may be implemented as a single program. One or more of the programs **126** may execute artificial intelligence (AI) algorithms. The programs **126** may include a data processing program **126**A that may be configured to process data in at least one of the instruments **104**, including the imaging instrument **104**B. For example, the data processing program **126**A may process data input to the system **100** and/or one or more of the instruments **104**. The data processing program **126**A may also process data generated by one or more of the instruments **104**.
(32) In some embodiments, the memory **124** may include a processing algorithm **126**B. In some embodiments, the memory **124** may include two or more processing algorithms. In some embodiments, the two or more processing algorithms may be implemented as a single processing algorithm, such as the processing algorithm **126**B. The processing algorithm **126**B may include computer code configured to analyze image data and other data generated by imaging devices in the system **100** or coupled to the system **100**. In some embodiments, the processing algorithm **126**B may process data using AI, which may be implemented as a tube characterization model **126**C that analyzes image data as described herein. The tube characterization model **126**C may be updated as described herein. For example, an image decomposition model **502** (FIG. **5**) and/or an image composition model **504** (FIG. **5**) may be updated as described herein.
(33) The imaging devices **216** and computers (e.g., computer **120**) used to process images captured by the imaging devices **216** may be part of machine vision and machine learning used by the system **100** to facilitate processing of the tube assemblies **102**. For example, as described herein, the machine vision and AI models that may be running in the tube characterization model **126**C may identify specific types of tube assemblies **102** and/or samples located in the tube assemblies **102** to facilitate sample processing and analyses as described herein.
(34) The computer **120** may be coupled to or implemented in a workstation **128**. The workstation **128** may enable user interaction with the system **100** and/or one or more of the instruments **104**. The workstation **128** may include a display **128**A and/or a keyboard **128**B that enable users to interface with the system **100** and/or individual ones of the instruments **104**.
(35) In some embodiments, the system **100** may be coupled to a laboratory information system (LIS) **130** that may determine how samples are to be tested by the system **100**. In some

embodiments, the LIS **130** may be implemented in the computer **120**. The LIS **130** may be coupled to a hospital information system (HIS) **132** that may receive specific assay orders for specific samples. The HIS **132** may also receive assay results after assays are performed on the specific samples by the system **100**.

(36) In an example of the operation of the system **100**, a doctor may order a specific test to be performed on a sample from a patient. The doctor may enter the test order into the HIS **132**. The HIS **132** may then transmit the test order to the LIS **130**. The sample may be drawn from the patient and placed into a tube assembly, such as the tube assembly **202** (FIG. **2**). The configuration of the tube assembly **202**, such as color, size, and shape, may correspond to the type of test in the test order. The tube assembly **202** may then be delivered to the system **100**. The LIS **130** may determine how the sample is to be tested and may transmit this information to the computer **120** where instructions may be generated to operate the instruments **104** to perform the test. For example, the data processing program **126**A may generate the instructions. The imaging devices **216** (FIG. **2**) may generate images of the tube assembly **202** and, based on the image data, the tube characterization model **126**C may identify the type of the tube assembly **202**. The testing and movement of the tube assembly **202** throughout the system **100** may be at least partially based on the type of tube assembly **202**. After testing is complete, the LIS **130** may transmit the test results to the HIS **132**.

(37) Additional reference is now made to FIGS. **3**A-**3**C, which illustrate different types of tube assemblies that may be used within the system **100**. Tube assemblies refer to tubes with or without caps attached to the tubes. Tube assemblies may also include samples or other contents of the tube assemblies. Additional reference is also made to FIGS. **4**A-**4**C, which illustrate the tube assemblies of FIGS. **3**A-**3**C without the caps. As shown in the figures, all the tube assemblies have different configurations or geometries. The caps may each have a different cap geometry.

(38) A tube assembly **340** of FIG. **3**A includes a cap **340**A that is white with a red stripe and has an extended vertical portion. The cap **340**A may fit over a tube **340**B. The tube assembly **340** has a height H**31**. FIG. **4**A illustrates the tube **340**B without the cap **340**A. The tube **340**B has a height H**41** and a width W**41**.

(39) A tube assembly **342** of FIG. **3**B includes a cap **342**A that is blue with a dome-shaped top. The cap **342**A may fit over a tube **342**B. The tube assembly **342** has a height H**32**. FIG. **4**B illustrates the tube **342**B without the cap **342**A. The tube **342**B has a height H**42** and a width W**42**.

(40) A tube assembly **344** of FIG. **3**C includes a cap **344**A that is red and gray with a flat top. The cap **344**A may fit over a tube **344**B. The tube assembly **344** has a height H**33**. FIG. **4**C illustrates the tube **344**B without the cap **344**A. The tube **344**B has a height H**43** and a width W**43**.

(41) FIGS. **3**A-**4**C illustrate different types of tubes, caps, and tube assemblies that may be used in the system **100**. The tube characterization model **126**C may have been trained to characterize or identify the tubes, caps, and tube assemblies of FIGS. **3**A-**4**C. In some embodiments, the tube characterization model **126**C may also determine whether the tube assemblies are not able to be characterized. In some embodiments, the tubes may have shapes other than the cylindrical shapes of FIGS. **3**A-**4**C and may be characterized by the tube characterization model **126**C. In addition, the tubes and caps may be made of different materials and may have different shapes, structures, albedo, and textures that may be characterized by the tube characterization model **126**C.

(42) Over time, laboratories may utilize tube assemblies from sources that provide tube assemblies that deviate from tube assemblies supplied by manufacturers on which the tube characterization model **126**C was trained. For example, the laboratories may use tube assemblies supplied by local manufacturers or new tube assembly designs that are not stored in the tube characterization model **126**C. The training data may be limited and may limit the number of tube variations the tube characterization model **126**C may identify or characterize. Retraining the tube characterization model **126**C to handle different tube types typically requires training with large numbers of images illustrating different configurations of tube assemblies under different views and lighting

conditions. In some conventional embodiments, the training is performed manually, which makes the retraining costs very high.

(43) The embodiments disclosed herein synthesize tube assembly images to generate synthetic tube assembly images. In one example, the tube characterization model **126**C is trained to characterize specific types of tube assemblies that may be used for specific tests. The tube assemblies may have specific tube materials, tube colors, tube shapes, cap shapes, cap colors, and other unique attributes. Over time, one or more manufacturers may change these attributes of the tube assemblies. The embodiments disclosed herein enable the system **100**, such as the processing algorithm **126**B, to synthesize images of tube assemblies with the new attributes, which may then be used to by the tube characterization model **126**C to characterize the new tube assemblies.

(44) In another example, the system **100** may receive a new tube assembly type from a new manufacturer. Each attribute of the new tube assembly type may be similar to the attributes of a specific tube assembly on which the tube characterization model **126**C has been trained. For example, the new tube assembly may have the same tube material as a tube assembly on which the tube characterization model **126**C has been trained but with a different cap shape. While another tube assembly type on which the tube characterization model **126**C was trained may have the same cap type as a new tube assembly type, but with a different tube material. The system **100**, using the tube characterization model **126**C, may synthesize images of the new tube assembly with the different cap shape. In some embodiments, the system **100** may synthesize the image using a similar cap shape from another type or tube assembly. In other embodiments, the system **100** may synthesize the cap shape of the new tube assembly.

(45) Some embodiments disclosed herein may use unpaired or paired image-to-image translation to adapt existing data (e.g., image data) from existing tube assembly types to new tube assembly types. The existing data may be the images used to train the tube characterization model **126**C. The image-to-image translation may be performed using machine learning, such as deep learning (DL) methods. An example of the machine learning includes generative adversarial networks (GANs). An example of a GAN is cycleGAN which performs unpaired image-to-image translation using cycle-consistent adversarial networks. Some embodiments decompose a captured image of a tube assembly into decomposed features. One or more of the decomposed features may be translated into a synthesized tube assembly image or into a portion of a synthesized tube assembly. Thus, the systems and methods disclosed herein enable collection of data (e.g., images) from existing tube assembly configurations to develop synthesized images of different tube assemblies.

(46) Additional reference is made to FIG. **5**, which illustrates a block diagram of a process **500** that implements a method of synthesizing images of tube assemblies as described herein. The images referred to in FIG. **5** may be image data, such as image data generated by the imaging instrument **104**B. The process **500** uses synthesized data or images of paired tube assemblies with controlled variations to train an image decomposition model **502** and an image composition model **504**. The trained image decomposition model **502** may then receive input images of tube assemblies and decompose the input images into latent space **506** having a plurality of features. A latent space manipulation module **508** may manipulate one or more features of the images in the latent space **506**. For example, the latent space manipulation module **508** may manipulate one or more features in the latent space that are related to an attribute such as the cap color of an image of a tube assembly. The manipulated features may then be used by the image composition model **504** to generate synthesized images of the tube assemblies.

(47) FIG. **6** illustrates a block diagram of a process **600** that implements a method of synthesizing images of tube assembly exterior surfaces and tube assembly interiors according to one or more embodiments. The tube assembly interior synthesis may synthesize fluids in the tube assemblies. In some embodiments, the process **600** may be considered two separate processes.

(48) The process **600** may include decomposing images using the image decomposition model **502** described in FIG. **5** that decomposes input images into various features. The process **600** may use a

tube exterior latent space **610** configured to characterize exterior attributes of images of the tube assemblies. The tube exterior latent space **610** may include individual variables or features **612** that may be configured to isolate different attributes of images of the tube assembly exteriors decomposed by the image decomposition model **502**. The features **612** may isolate the images based on attributes such as geometries and materials of the tube assemblies, surface properties of the tube assemblies, lighting conditions under which the images are captured, and camera properties of imaging devices (e.g., imaging devices **216** of FIG. **2**). Other attributes of the images may be isolated or characterized.

(49) The tube exterior latent space **610** may be manipulated by a latent space manipulation module **614**. The latent space manipulation module **614** may be identical or substantially similar to the latent space manipulation module **508** of FIG. **5**. The latent space manipulation module **614** may manipulate features **612** isolated in the tube exterior latent space **610**. The image composition model **504** may then compose the image using the manipulated features manipulated by the latent space manipulation module **614**.

(50) The process **600** may include decomposing images of tube interiors or fluids using the image decomposition model **502** described in FIG. **5**. The process **600** may use a tube interior latent space **618** configured to synthesize interior attributes of images of the tube assemblies. In some embodiments, the tube interior latent space **618** may characterize contents of the tube assemblies, such as samples stored in the tube assemblies, into features **620**. The tube interior latent space **618** may include individual variables or features **620** that may include geometries and materials of the tube assemblies, fluid properties, lighting conditions under which the images are captured, camera properties of imaging devices (e.g., imaging devices **216** of FIG. **2**), and label and/or barcode properties. Other attributes of the images may be isolated or characterized into the features **620**.

(51) The features **620** of the tube interior latent space **618** may be manipulated by a latent space manipulation module **624**. The latent space manipulation module **624** may be identical or substantially similar to the latent space manipulation module **614** or the latent space manipulation module **508** of FIG. **5**. The latent space manipulation module **624** may manipulate features **620** isolated in the tube interior latent space **618**. The image composition model **504** may then compose synthesized images of tube assemblies using the manipulated features.

(52) Additional reference is made to FIG. **7**, which is an embodiment of a training process **700** that may be used to train the image decomposition model **502** and the image composition model **504**. A first image A of a first tube assembly is input to an encoder network that is implemented as a decomposition model **730** (sometimes referred to as an image decomposition model) that generates latent features in a latent variable LA in a multidimensional latent space. The decomposition model **730** may be similar or identical to the decomposition model **530**. In some embodiments, the decomposition model **730** generates latent features in low-dimensional space. The features of the latent variable LA may be partitioned into multiple groups of features such as W, X, Y, and Z as shown in FIG. **7**. Each of the features may correspond to a specific one of the attributes of a tube assembly. In some embodiments, at least one of the features (e.g., feature Z) may be reserved for an intrinsic property of images of the tube assemblies in which variations cannot be controlled by a setup. In some examples, all tubes in a study may have a cylindrical structure that all share in common such that it is expected that they all share the same values in feature Z.

(53) A composition model **732** (sometimes referred to as an image composition model) that may be implemented as a decoder network may use the features of the latent variable LA to reconstruct an image A′ of a tube assembly to be as similar to the input image A as possible. The composition model **732** may be similar or identical to the composition model **532**. A reconstruction loss module **734** may compare the input image A to the output image A′ to determine whether the images are close to each other. For example, the reconstruction loss module **734** may analyze the attributes of the image A and the image A′ to determine whether the latent variable LA is correct and whether the decomposition model **730** and the composition model **732** are trained correctly.

(54) The training process **700** may also include receiving a second input image B of a second tube assembly into the decomposition model **730** and generating latent features in a latent variable LB in multidimensional space. The images A and B may be paired tube assembly images. The features of the latent variable LB may be partitioned into multiple groups of features such as W′, X′, Y′, and Z′ as shown in FIG. **7**. Each of the features may correspond to a specific one of the attributes of the tube assemblies. At least one of the features (e.g., feature Z′) may be reserved for the intrinsic property of the tube image that cannot be captured by one of the features. The composition model **732** may use the features to reconstruct an image B′ of a tube assembly to be as similar to the input image B as possible. A reconstruction loss module **736** may compare the input image B to the output image B′ to determine whether the images are close to each other. For example, the reconstruction loss module **736** may analyze the attributes of the image B and the image B′ to determine whether the latent variable LB is correct and whether the decomposition model **730** and the composition model **732** are trained correctly.
(55) Training a disentangled latent space, such as used with the decomposition model **730** and the composition model **732**, may include having the input image B be a specific variation relative to the input image A. For example, in the case of tube cap synthesis, a first feature (e.g., W and W) of the latent variable LA and the latent variable LB may be the albedo such as the color of tube caps. In this example, the input image B is the same tube assembly type as the input image A with the only variation being color of the tube cap. The input image B may either be generated synthetically such as generated using computer-generated imagery (CGI) or captured from the same type of tube assembly and the same imaging setup used to acquire the input image A. The tube cap in the input image B may be very similar to the cap in the input image A, with the only or only significant difference being a difference in cap colors. In the latent spaces LA and LB, the remaining features (e.g., X and X′, Y and Y′, Z and Z′) are to be as similar as possible while allowing the first feature W and W to vary as described above.
(56) The above process may be repeated for other variations in the latent spaces LA and LB, so that each of the features of the latent variables captures a particular variation. In some embodiments, it is possible to have multiple variations at the same time between input image A and input image B. In such embodiments, corresponding latent features may be allowed to vary while the remaining features are held consistent. As an example, input image A and input image B may be captured with the same camera, but include tube assemblies having different tube geometries, surface properties, and under different lighting conditions. In this embodiment, only the latent feature corresponding to camera properties are forced to be similar and the other features may vary.
(57) The architecture of the training process **700** includes the decomposition model **730** and the composition model **732**. Therefore, the training may be enhanced with unpaired images by training on those images to minimize the reconstruction loss only. Such images can also be edited in a latent domain by replacing some of their latent features with latent features taken from another image. Then the decomposition model **730** and the composition model **732** can be trained by an adversarial network trying to classify the edited and unedited images. For example, there may be a set of tube assembly images S.sub.1 captured by a first imaging device C.sub.1 (e.g., the first imaging device **216**A of FIG. **2**) with lighting condition L.sub.1 provided by the first illumination device **217**A. Latent features of images S.sub.1 corresponding to camera properties of the first imaging device **216**A and light conditions provided by the first illumination device **217**A may be similar to each other. The two latent features may be regularized to approximate a unit Gaussian distribution with mean vectors (m.sub.c1, m.sub.L1). For another set of tube assembly images Sz captured by a second imaging device C.sub.2 (e.g., the second imaging device **216**B) with lighting condition L.sub.2 provided by the second illumination device **217**B, latent features related to the camera and lighting conditions are forced to approximate a unit Gaussian distribution with mean vectors (m.sub.c2, m.sub.L2).
(58) By replacing the latent features of a tube image in S.sub.1 with vectors (m.sub.c2, m.sub.L2)

and reconstructing the image of the tube assembly using the composition model **732**, the same tube captured with C.sub.2 (the second imaging device **216**B) and L.sub.2 (second illumination device **217**B) will be synthesized. In some embodiments, an adversarial discriminator may be further trained to enforce rules wherein the synthesized image is to be indistinguishable from images in S.sub.2 in terms of the camera and the lighting. Similarly, the latent features of a tube assembly image in S.sub.2 with (m.sub.c1, m.sub.L1) may be replaced to synthesize images captured with C.sub.1 and L.sub.1. These processes ensure that image variations are disentangled by the latent space (e.g., latent space **610** and tube interior latent space **618**). After training, the trained image decomposition model **730** learns how to perform image decomposition into a disentangled latent space, and the trained image composition model **732** learns how to perform image composition from a latent variable, such as one of the latent variables L.sub.A or L.sub.B, in the latent spaces. Since each latent space is disentangled using known variations, a latent vector of a given tube assembly image can be perturbed to generate numerous tube assembly images with desired variations.

(59) In some embodiments, the latent space representation may encode data (e.g., decompose images) across different scales. For example, normalizing flows, autoregressive models, variational autoencoders (VAEs), and deep energy-based models are examples for deep generative learning. Furthermore, the latent space may be constrained to be a known parametric distribution (e.g., Gaussian or mixture-of-Gaussian) or a non-parametric distribution, such as with a vector quantized variational autoencoder (VQ-VAE).

(60) In addition, images other than the generated images can be used as image data to characterize the attributes, such as tube cap classification and tube fluid characterization. The disentangled features in the latent space may also provide useful information (either used as ground-truth annotation or treated as discriminative features) for downstream tasks that may be performed by the system **100**. Furthermore, the disentangled features in the latent space can also be used to develop tools that can be used by laboratory personnel, such as vendors, research and development staff, field service technicians, and/or end users (e.g., lab technicians and managers) to evaluate new tube assembly types and tube assembly variations to determine the ability of the system **100** (FIG. **1**) to properly characterize the tube assemblies.

(61) In some embodiments, the system **100** may use unpaired image-to-image translation such as cycleGAN to adapt existing data from images of old tube assemblies to images of new tube assemblies. This procedure enables developers of the system **100** and/or tube assemblies to collect field data from the existing tube assemblies when developing new types of tube assemblies.

(62) In some embodiments, image data may be collected in groups that may include dense collections of representative tube assembly types and sparse collections of other similar tube assembly types. Augmented data may be used to fill in missing attributes of images of the sparsely collected tube assembly types. For example, color augmentation may be used to fill in missing colors in the images of sparsely collected tube assembly types.

(63) An example of the system **100** synthesizing a tube assembly is described below. Reference is made to FIG. **8**A and FIG. **8**B, which illustrate images of similar tube assemblies. The tube assembly **840** of FIG. **8**A is identical or similar to the tube assembly **842** of FIG. **8**B except that the color of the cap **840**A of the tube assembly **840** is blue and the color of the cap **842**A is green. For example, the tube **840**B may be similar to the tube **842**B and both the tube assembly **840** and the tube assembly **842** may have the same height H**81**. The decomposition model **730** may receive an image of the tube assembly **840** as input image A. The decomposition model **730** may also receive an image of the tube assembly **842** as input image B. The decomposition model **730** decomposes the input image A into a set of features (W, X, Y, Z) in the latent space represented as a variable L.sub.A and the image B into a set of features (W′, X′, Y′, Z′) in the same latent space represented as a variable L.sub.B.

(64) Additional reference is made to FIG. **8**C, which provides an example of a decomposed image

844 of the tube assembly **840**. It is noted that other decompositions of the image of the tube assembly **840** may be performed. In the embodiment of FIG. **8**C, the feature W may be the cap color, which is blue. The feature X may be the cap height H**82**, the feature Y may be the tube height H**83**, and the feature Z may be the tube width W**81**. Other features may include other portions of the tube geometry or the tube assembly geometry. The composition model **732** may reconstruct the deconstructed image to generate the image A′. The input image A may be compared to the reconstructed image A′ by the reconstruction loss module **734**. If the loss or difference between the input image A and the reconstructed image A′ is within a predetermined amount, the decomposition model **730** and composition model **732** may be considered properly trained and the latent variable L.sub.A is correct for processing the tube assembly **840**.
(65) The tube assembly **842** may be selected as a paired image because the only significant difference between the tube assembly **840** and the tube assembly **842** is the color of the caps. The decomposition model **730** may decompose the image B of the tube assembly **842** per the latent space L.sub.B. The features of the latent space L.sub.B are the same as with the latent space L.sub.A. The feature W′ may be the cap color, which is green, the feature X′ may be the cap height H**82**, the feature Y may be the tube height H**83**, and the feature Z may be the tube width W**81**. The composition model **732** may compose the deconstructed image to generate the image B′. The input image B may be compared to the reconstructed image B′ by the reconstruction loss module **736**. If the loss or difference between the input image B and the reconstructed image B′ is within a predetermined amount, the decomposition model **730** and composition model **732** may be considered properly trained and the latent variable L.sub.B is correct for processing the tube assembly **842**.
(66) The green cap **842**A may be a new cap color. In order to reconstruct the tube assembly **842** with a blue cap, the feature W, which is the green cap is substituted with the feature W. The image-to-image reconstruction and other methods may be used to reconstruct the tube assembly **842** with the blue cap. The resulting reconstructed or synthesized image using features W, X′, Y′, and Z′ will be the tube assembly **842**A with the blue cap of the tube assembly **840**. In some embodiments, the cap **840**A as a whole may be used in place of the cap **842**A and in other embodiments, the color of the cap **840**A may be used in place of the color of the cap **842**A. In other embodiments, the color of the cap **840**A may be stored and used for the cap **842**A without imaging the tube assembly **840**.
(67) Reference is now made to FIGS. **9**A-**9**B, which illustrate tube assemblies having different interior and/or fluid features that may be used with tube interior latent space **618** (FIG. **6**). The tube assembly **940** may include one or more items (e.g., attributes) located in the interior of the tube **940**B and may be sealed by a cap **940**A. One or more fluids **946** may be located in the interior of the tube **940**B and may include a blood serum **946**A, a gel separator **946**B, and blood **946**C. An air gap **946**D may be located above the fluids **946**.
(68) The tube assembly **942** may include one or more items (e.g., attributes) located in the interior of the tube **942**B and may be sealed by a cap **942**A. One or more fluids **948** may be located in the interior of the tube **942**B and may include blood serum **948**A, a gel separator **948**B, and blood **948**C. An air gap **948**D may be located above the fluids **948**. The material of the tube **940**B and the material of the tube **942**B may be similar and may be identified by generating image data through the air gap **946**D and the air gap **948**D. Accordingly, the attributes of the tube assembly **940** and the tube assembly **942** include, but are not limited to, tube geometry, tube material, tube color, tube surface properties, sample fluid in the tubes, labels affixed to the tube, and lighting condition of the tube during image capture.
(69) All the above attributes of the tube assembly **940** and the tube assembly **942** may be identical or substantially similar. The difference between the tube assembly **940** and the tube assembly **942** may be labels attached to the tube **940**B and the tube **942**B. The tube assembly **940** has a label **950** affixed to the tube **940**B and the tube assembly **942** has a label **952** affixed to the tube **942**B. As shown in FIGS. **9**A-**9**B, the label **950** differs from the label **952**. For example, the size of the label

950 differs from the size of the label **952** and the position of a barcode **954** differs from the location of a barcode **956**.

(70) The decomposition model **730** and the composition model **732** may be configured to identify and/or characterize the attributes of the images of the tube assembly **940** and the tube assembly **942** into features in latent space. Accordingly, the system **100** may be configured to substitute the above features into synthesized images (e.g., image A′ or B′). In the example of FIGS. **9**A-**9**B, the system **100** may substitute labels from an original image for either of the label **950** or the label **952** to generate a synthesized image. In some embodiments, the system **100** may substitute labels, but keep the original bar code or other label information in the synthesized image.

(71) In some embodiments, the synthesized tube assembly images are used by the system **100** during a tube assembly identification process. For example, when tube assemblies **102** are loaded into the input/output instrument **104**A, images of the tube assemblies **102** may be captured by the imaging device **116** or another imaging device. Images of synthesized tube assemblies may be used for tube assembly identification of tube assemblies that were not initially recognized by the tube characterization model **126**C. The tube assemblies may then be characterized and transported to the appropriate instruments **102** where the samples are analyzed as described herein. In some embodiments, the instruments **102** may perform processes, such as aspiration, that are customized to the identified tube assemblies. In other embodiments, the robot **108** (FIG. **1**) may move sample containers to and from carriers **114** using specific techniques, such as specific grasping techniques, depending on the type of identified tube assemblies.

(72) Reference is now made to FIG. **10**, which is a flowchart illustrating a method **1000** of synthesizing an image of a tube assembly (e.g., tube assembly **202**). The method **1000** includes, at **1002**, capturing an image of a tube assembly, the capturing generating a captured image. The method **1000** includes, at **1004**, decomposing the captured image into a plurality of features (e.g., features **612**) in latent space (e.g., latent space **610**) using a trained image decomposition model (e.g., image decomposition model **502**). The method **1000** includes, at **1006**, manipulating one or more of the features in the latent space into manipulated features. For example, a cap color may be changed from red to blue and/or a cap shape may be changed from round to oval. Other features may be additionally or alternatively changed (i.e., manipulated). The method **1000** includes, at **1008**, generating a synthesized tube assembly image with at least one of the manipulated features using the trained image composition model (e.g., image composition model **504**).

(73) Reference is now made to FIG. **11**, which is a flowchart illustrating a method **1100** of synthesizing images of tube assemblies (e.g., tube assembly **202**). The method **1100** includes, at **1102**, constructing an image decomposition model (e.g., image decomposition model **502**) configured to receive input images of tube assemblies and to decompose the input images into a plurality of features (e.g., features **612**) in a latent space (e.g., latent space **610**). The method **1100** includes, at **1104**, constructing an image composition model (e.g., image composition model **504**) configured to compose synthetic tube assembly images based on the plurality of features in the latent space. The method **1100** includes, at **1106**, training the image decomposition model and the image composition model using at least an image of a first tube assembly (e.g., tube assembly **840**) and an image of a second tube assembly (e.g., tube assembly **842**) with one or more known variations between the image of the first tube assembly and the image of the second tube assembly, wherein the training produces a trained image decomposition model and a trained image composition model.

(74) While the disclosure is susceptible to various modifications and alternative forms, specific method and apparatus embodiments have been shown by way of example in the drawings and are described in detail herein. It should be understood, however, that the particular methods and apparatus disclosed herein are not intended to limit the disclosure but, to the contrary, to cover all modifications, equivalents, and alternatives falling within the scope of the claims.

Claims

1. A method of synthesizing an image of a tube assembly, comprising: capturing an image of a tube assembly, the capturing generating a captured image; decomposing the captured image into a plurality of features in latent space using a trained image decomposition model; manipulating one or more of the features in the latent space into manipulated features; and generating a synthesized tube assembly image with at least one of the manipulated features using a trained image composition model.
2. The method of claim 1, further comprising training an image decomposition model and an image composition model using at least an image of a first tube assembly and an image of a second tube assembly, wherein there are one or more known variations between the image of the first tube assembly and the image of the second tube assembly, the training producing the trained mage image decomposition model and the trained image composition model.
3. The method of claim 2, wherein the training the decomposition model is at least partially based on a reconstruction loss between synthesized images and input images.
4. The method of claim 1, wherein the tube assembly comprises a tube and a cap and at least one of the plurality of features is tube geometry, tube color, tube surface property, cap geometry, cap color, or lighting condition.
5. The method of claim 1, wherein the tube assembly comprises a tube and at least one of the plurality of features is tube geometry, tube material, sample fluid in the tube, label affixed to the tube, or lighting condition of the tube.
6. A method of synthesizing images of tube assemblies, comprising: constructing an image decomposition model configured to receive input images of tube assemblies and to decompose the input images into a plurality of features in a latent space; constructing an image composition model configured to compose synthetic tube assembly images based on the plurality of features in the latent space; and training the image decomposition model and the image composition model using at least an image of a first tube assembly and an image of a second tube assembly with one or more known variations between the image of the first tube assembly and the image of the second tube assembly, wherein the training produces a trained image decomposition model and a trained image composition model.
7. The method of claim 6, further comprising: capturing an image of a tube assembly, the capturing generating a captured image; decomposing the captured image into a plurality of decomposed features in the latent space using the trained image decomposition model; manipulating one or more of the decomposed features in the latent space into manipulated features; and generating a synthesized tube assembly image with at least one of the manipulated features using the trained image composition model.
8. The method of claim 6, wherein the training the decomposition model is at least partially based on a reconstruction loss between synthesized images and input images.
9. The method of claim 6, wherein the image composition model is trained based on a reconstruction loss between synthesized images and input images.
10. The method of claim 6, wherein at least one of the features in the latent space is generated using computer-generated imagery.
11. The method of claim 6, wherein images of at least two of the tube assemblies differ from each other with one or more controlled attributes.
12. The method of claim 6, wherein images of at least two tube assemblies share one or more attributes in common.
13. The method of claim 6, wherein at least one of the image decomposition model or the image composition model is trained based on a reconstruction loss without the images of the tube assemblies.

14. The method of claim 6, wherein the tube assemblies comprise a tube and a cap and at least one of the plurality of features in the latent space is tube geometry, tube color, tube surface property, cap color, cap geometry, or lighting condition.

15. The method of claim 6, wherein the tube assemblies comprise a tube and at least one of the plurality of features in the latent space is tube geometry, tube material, sample fluid in the tube, label affixed to the tube, or lighting condition of the tube.

16. The method of claim 6, wherein one or more features in the latent space include a fluid property in a tube assembly.

17. A diagnostic laboratory system, comprising: an image decomposition model configured to receive input images of tube assemblies and to decompose the input images into a plurality of features in a latent space; and an image composition model configured to compose synthetic tube assembly images based on the plurality of features in the latent space, wherein the image decomposition model and the image composition model are trained using at least an image of a first tube assembly and an image of a second tube assembly with one or more known variations between the image of the first tube assembly and the image of the second tube assembly.

18. The diagnostic laboratory system of claim 17, wherein the image decomposition model is trained at least partially based on a reconstruction loss between synthesized images and input images.

19. The diagnostic laboratory system of claim 17, wherein the image composition model is trained based on a reconstruction loss between synthesized images and input images.

20. The diagnostic laboratory system of claim 17, wherein the tube assemblies comprise a tube and a cap and at least one of the plurality of features in the latent space is tube geometry, tube color, tube surface property, cap color, cap geometry, or lighting condition.